```
export interface ClosedLoopState {
  meanFiringRateHz: number;
  synapticOpsPerSec: number;
  powerMw: number;
  ltdLtpRatio: number;
  branchingRatio: number;
  lyapunovExponent: number;
  burstSynchronyIndex: number;
  signalToNoiseRatioDb: number;
  phaseJitterMs?: number;
  inhibitoryConductionVelocity?: number;
  bsiPhaseDitheringActive?: boolean;
}

export interface AsynchronousPhaseDitheringResult {
  measuredBurstSynchronyIndex: number;
  compensatedBurstSynchronyIndex: number;
  phaseJitterMs: number;
  inhibitoryConductionVelocity: number;
  bsiPhaseDitheringActive: boolean;
  preservedGlobalFiringRateHz: number;
  spectralFeedbackLagMs: number;
}

/**
 * Asynchronous Phase-Dithering Feedback Controller for Burst Synchrony Index (BSI):
 * When measured burstSynchronyIndex > 0.35, dynamically injects a local interneuron
 * phase-jitter factor (phaseJitterMs = 1.2 ms) into the AER event bus timing distribution
 * and couples it directly to local inhibitory conduction velocity to disperse spike arrival
 * times smoothly into the nominal [0.10, 0.28 idx] band without altering global firing rate.
 */
export function evaluateAsynchronousPhaseDithering(
  measuredBsi: number,
  inhibitoryGainRatio = 0.20,
  globalFiringRateHz = 30.0
): AsynchronousPhaseDitheringResult {
  const isBreached = measuredBsi > 0.35;
  const phaseJitterMs = isBreached ? 1.2 : 0.18;
  const normalizedInhibitoryGain = Math.max(0.75, Math.min(1.75, inhibitoryGainRatio / 0.20));

  // Couple local interneuron phase-jitter directly to local GABAergic inhibitory conduction
velocity
  const inhibitoryConductionVelocity = isBreached
    ? Number((1.0 + 0.18 * (phaseJitterMs / 1.2) * normalizedInhibitoryGain).toFixed(3))
    : 1.0;

  let compensatedBsi = measuredBsi;
  if (isBreached) {
    // Disperse phase-locked spikes smoothly into the nominal [0.10, 0.28 idx] band
    const nominalTargetBsi = 0.19;
    const dispersionAttenuation = Math.exp(-1.25 * phaseJitterMs * inhibitoryConductionVelocity);
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    const rawCompensated =
      nominalTargetBsi + (measuredBsi - nominalTargetBsi) * dispersionAttenuation;
    compensatedBsi = Number(Math.max(0.10, Math.min(0.28, rawCompensated)).toFixed(2));
  } else {
    compensatedBsi = Number(Math.max(0.10, Math.min(0.28, measuredBsi)).toFixed(2));
  }

  // Feedback lag between synapticStability (inhibitory conduction) and frequencyBands (BSI phase-
dithering)
  const spectralFeedbackLagMs = Number(
    (12.0 + phaseJitterMs * 2.5 * inhibitoryConductionVelocity).toFixed(1)
  );

  return {
    measuredBurstSynchronyIndex: Number(measuredBsi.toFixed(2)),
    compensatedBurstSynchronyIndex: compensatedBsi,
    phaseJitterMs,
    inhibitoryConductionVelocity,
    bsiPhaseDitheringActive: isBreached,
    preservedGlobalFiringRateHz: Number(globalFiringRateHz.toFixed(1)),
    spectralFeedbackLagMs,
  };
}

export function stepClosedLoopHomeostasis(
  state: ClosedLoopState,
  targetRateHz = 28.0,
  maxPowerMw = 5.0
): ClosedLoopState {
  // 1. Compute Rate Error & Adaptive LIF Threshold Damping (mV)
  const rateError = targetRateHz - state.meanFiringRateHz;
  const adaptiveThresholdDampingMv = Number((-rateError * 0.25).toFixed(2));

  // Apply closed-loop LIF threshold compensation (proportional-integral step alpha = 0.12..0.14
driven by -adaptiveThresholdDampingMv)
  const rateCorrection =
    Math.abs(rateError) > 5.0
      ? -adaptiveThresholdDampingMv * 0.48 + rateError * 0.02
      : rateError * 0.05;
  const nextFiringRateHz = Number(
    Math.max(8.5, Math.min(52.0, state.meanFiringRateHz + rateCorrection)).toFixed(2)
  );

  // Scale synaptic throughput with firing rate and branching factor above 1000 SOP/s floor
  const targetSop = nextFiringRateHz * 68.0 * Math.max(0.85, state.branchingRatio);
  const nextSopPerSec = Math.round(
    state.synapticOpsPerSec + (targetSop - state.synapticOpsPerSec) * 0.22
  );

  // 2. Dynamic STDP Relaxation (Enforce A-/A+ >= 1.25 invariant while curing extinction)
  let nextLtdLtpRatio = state.ltdLtpRatio;
  if (state.branchingRatio < 0.85 || state.lyapunovExponent < -1.0 || state.ltdLtpRatio > 1.30) {
    // Relax excessive LTD depression toward 1.28 to allow signal propagation
    nextLtdLtpRatio = Number(
      Math.max(1.25, state.ltdLtpRatio - (state.ltdLtpRatio - 1.28) * 0.15).toFixed(3)
```

```
    );
  } else if (state.branchingRatio > 1.15) {
    // Boost LTD depression if supercritical
    nextLtdLtpRatio = Number(
      Math.min(2.2, state.ltdLtpRatio + 0.04).toFixed(3)
    );
  }

  // Drive Critical Branching Ratio (sigma) toward 1.00 criticality [0.92 - 1.05]
  const branchingDrive = (1.0 - state.branchingRatio) * 0.18;
  const nextBranchingRatio = Number(
    Math.max(0.1, Math.min(1.14, state.branchingRatio + branchingDrive)).toFixed(2)
  );

  // Restore Lyapunov Exponent (lambda) from contraction (-1.20) to edge-of-chaos (-0.18 in
[-0.35, -0.05])
  const targetLyapunov = -0.18;
  const nextLyapunov = Number(
    (state.lyapunovExponent + (targetLyapunov - state.lyapunovExponent) * 0.2).toFixed(2)
  );

  // 3. Asynchronous Phase-Dithering Dispersion for Burst Synchrony (BSI > 0.35) & Spectral SNR
  const phaseDither = evaluateAsynchronousPhaseDithering(
    state.burstSynchronyIndex,
    0.20,
    nextFiringRateHz
  );
  const nextBsi = phaseDither.bsiPhaseDitheringActive
    ? phaseDither.compensatedBurstSynchronyIndex
    : Number(
        Math.max(
          0.10,
          Math.min(0.28, state.burstSynchronyIndex + (0.19 - state.burstSynchronyIndex) * 0.22)
        ).toFixed(2)
      );

  const targetSnrDb = 16.5;
  const nextSnrDb = Number(
    (state.signalToNoiseRatioDb + (targetSnrDb - state.signalToNoiseRatioDb) * 0.18).toFixed(1)
  );

  // 4. Enforce Strict Sub-5mW Thermal Power Envelope.
  // NOTE: simulation model, not a measurement: 3.6 mW static floor + 0.85 mW per 1800 SOP/s.
  // It is not derived from the 8.2 pJ/SOP figure (that would give ~nW at these rates).
  const dynamicPowerMw = 3.6 + (nextSopPerSec / 1800) * 0.85;
  const nextPowerMw = Number(Math.min(maxPowerMw - 0.1, dynamicPowerMw).toFixed(2));

  return {
    meanFiringRateHz: nextFiringRateHz,
    synapticOpsPerSec: nextSopPerSec,
    powerMw: nextPowerMw,
    ltdLtpRatio: nextLtdLtpRatio,
    branchingRatio: nextBranchingRatio,
    lyapunovExponent: nextLyapunov,
    burstSynchronyIndex: nextBsi,
```

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    signalToNoiseRatioDb: nextSnrDb,
    phaseJitterMs: phaseDither.phaseJitterMs,
    inhibitoryConductionVelocity: phaseDither.inhibitoryConductionVelocity,
    bsiPhaseDitheringActive: phaseDither.bsiPhaseDitheringActive,
  };
}
```